PHYS 253: Thermal Physics

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Abstract

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1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. Grading is divided as follows:

- 10% problem sets (~ 6, submitted as handwritten PDF scans)[1]
- 50% two midterms
- 40% final exam

2 Thermodynamics

August 31 - September 2, 2026

Definition 1 (Thermodynamics)**.** *Thermodynamics, from the ancient Greek therme (heat) and dynamis (power), is the branch of physics that deals with the relationships between heat, work, temperature and energy. It studies the equilibrium properties of large-scale systems in which temperature is an important variable.*

Remark 1. *Thermodynamics does not describe the microscopic details of a system*[2]*. It describes the macroscopic properties and behaviour of a system using a small number of state variables.*

2.1 Terminologies

Definition 2 (System, surroundings, boundary)**.** *The system is everything involved in the process under study; the surroundings are the rest of the universe it interacts with; the boundary is the interface between them.*

Definition 3 (Open, closed, isolated)**.** *An open system exchanges matter and energy with its surroundings, a closed system only energy, and an isolated system neither. The boundary determines which.*

2.2 Equilibrium States

Definition 4 (Equilibrium state)**.** *A system is in an equilibrium state if all the measurable bulk physical properties of the system are uniform throughout the system and constant in time.*

Remark 2. *For a fixed volume of gas in an equilibrium state, the pressure is the same everywhere in the gas and the volume does not change in time.*

Remark 3. *The pressure, volume and mass of a gas fully describe its equilibrium state: once those are fixed, every other measurable bulk property is fixed as well.*

[1] Only one or two problems on each assignment are marked for submission and grading.
[2] Statistical mechanics, and more recent extensions such as stochastic thermodynamics, do.

Definition 5 (Local thermodynamic equilibrium)**.** *Divide the system into small volume elements. A volume element V_i is in local thermodynamic equilibrium if the properties inside V_i are the same as those of an equilibrium system at the local (P, V, T) conditions.*

Remark 4. *This lets a system with, say, a temperature gradient still be treated cell by cell, with U, H, S defined in each cell. The approximation is better when relaxation is slow and the gradients in (P, V, T) are gentle.*

Remark 5. *Thermodynamics studies equilibrium states themselves, not how long it takes to reach them.* [3]

2.3 Equations of State and State Functions

Definition 6 (Equation of state)**.** *An equation of state is a functional relationship between the easily measured state variables*[4] *of a system in an equilibrium state.*

Law 1 (Ideal gas law)**.** *For a dilute gas at high temperature in an equilibrium state,*

$$PV = nRT = Nk_BT, \tag{1}$$

where n is the number of moles, $N = nN_A$ is the number of molecules, R is the ideal gas constant and $k_B = R/N_A = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ is Boltzmann's constant.

Example. *The ideal gas law is an equation of state relating (P, V, T).*

Definition 7 (State function)**.** *A state function is a property of a system that depends only on the current equilibrium state, and not on the path taken to reach that state. Internal energy, enthalpy, entropy and Gibbs free energy are all state functions.*

Remark 6. *State functions matter because their values can be compared between two equilibrium states without knowing anything about the process that connected them.*

Example. *Temperature is a state function. It is the property that determines whether two systems are in thermal equilibrium*[5] *with each other.*

3 Laws of Thermodynamics

3.1 Zeroth Law

Law 2 (Zeroth Law of Thermodynamics)**.** *If two systems A and B are each separately in thermal equilibrium with a third system C, meaning no heat flows*

[3] Rates of processes belong to a separate subject (transport theory or kinetics).

[4] Such as pressure, volume and temperature.

[5] Thermal equilibrium is not full thermodynamic equilibrium, which also requires mechanical and chemical (diffusive) equilibrium.

between A and C and no heat flows between B and C, then A and B are in thermal equilibrium with each other.

Remark 7. *The zeroth law is what makes a thermometer work. System C is the thermometer, and the law guarantees that two systems reading the same value on it are in thermal equilibrium with each other.*

Remark 8. *Extrapolating $V(T)$ at fixed P for any sample of ideal gas gives $V \to 0$ at the same temperature, $-273.15\,^\circ C = 0$ K, which defines absolute zero. A gas thermometer reads $T = PV/nR$.*

3.2 First Law

Law 3 (First Law of Thermodynamics)**.** *The relationship between the heat Q added to a system, the work W done on it and the change in its internal energy U is dictated by the conservation of energy:*

$$\Delta U = Q + W. \tag{2}$$

Remark 9 (Sign Convention)**.** *$Q > 0$ when heat flows into the system. $W > 0$ when work is done on the system. Both raise the internal energy.*

Remark 10. *For an isolated system $Q = W = 0$, so U is constant.*

Remark 11. *The first law says energy is conserved. The second law (Section 10) says in which direction spontaneous processes go.*

4 Energy, Work and Heat

September 4, 2026

4.1 Energy

Definition 8 (Energy)**.** *Energy, from the Greek energeia meaning "in work," is the capacity to do work.*

Definition 9 (Kinetic energy)**.** *From the Greek kinesis (motion), the energy associated with the motion of a body or of the particles inside it.*

Definition 10 (Potential energy)**.** *Energy due to condition, position or composition. It is associated with interactions, meaning forces of attraction or repulsion between objects, or chemical bonding.*

Definition 11 (Internal energy of an ideal gas)**.** *For an ideal gas the internal energy depends only on temperature. It is the number of molecules times the average kinetic energy per molecule:*

$$U(T) = N\bar{e}_k = \frac{f}{2} N k_B T, \tag{3}$$

where f is the number of degrees of freedom per molecule. For a monatomic gas $f = 3$ and $f = 5$ for diatomic, so $U = \frac{3}{2} N k_B T$. More complex molecules have extra rotational or vibrational degrees of freedom, so $f > 3$.

4.2 Work

Definition 12 (Work)**.** *Work is done when a force acts through a distance, and it is associated with a change in the energy of a system:*

$$W = \int \vec{F} \cdot d\vec{r}.$$

In thermodynamics, work is any transfer of energy into or out of a system that is not heat.

Definition 13 (Quasistatic Process)**.** *Idealized thermodynamic process that happens slowly enough for a system to stay in continuous internal equilibrium.*

Compression of a gas by a piston. Take a gas in a cylinder closed by a piston of area A. Pushing the piston in by δx does work $W = F\,\delta x = PA\,\delta x$. For a quasistatic compression[6], $\delta V = -A\,\delta x$, so the work done *on* the gas is

$$W = -P\,\delta V \qquad \text{(quasistatic)}.$$

Compressing the gas means $\delta V < 0$, so $W > 0$ and the energy of the gas goes up. For a finite change,

$$W = -\int_{V_i}^{V_f} P\,dV, \tag{4}$$

which is the area under the curve on a P-V diagram.

Remark 12. *Energy and work are measured in joules,* $1\,\mathrm{J} = 1\,\mathrm{N\,m} = 1\,\mathrm{kg\,m^2\,s^{-2}}$.

4.3 Heat

Definition 14 (Heat)**.** *Heat Q is energy that flows between a system and its surroundings because of a temperature difference. It flows spontaneously from the warmer body to the colder one, by conduction, convection or radiation, until $T_A = T_B$. No temperature difference, no heat flow.*

Remark 13. *Heat and work are not properties of a system. They are the two ways energy can be exchanged between a system and its surroundings. A system does not "contain" heat or work, it contains energy.*

5 Reversible Processes and Differentials

5.1 Reversible and Irreversible Processes

Definition 15 (Thermodynamic process)**.** *A process is the mechanism that takes a system from one equilibrium state to another. The two equilibrium states are its end points.*

[6] Slow enough that the gas stays in internal equilibrium at every instant, so the pressure is uniform throughout the volume and the equation of state still applies.

Definition 16 (Reversible process)**.** *A process is reversible (quasistatic) if the system passes through a succession of equilibrium states, so the equation of state holds at every point, and at every step the system and surroundings could be returned to their original states. This requires*

- *reversible work: mechanical equilibrium, $P_{ext} = P_{sys}$;*
- *reversible heat: thermal equilibrium, $T_{surr} = T_{sys}$.*

Remark 14. *Reversible processes are an idealization: nothing would actually happen unless there were at least a tiny difference dP or dT between system and surroundings. They are still useful: a state function's change depends only on the end points, so it can always be computed along a reversible path.*

Example (Isothermal expansion of an ideal gas, $V_i \to V_f$)**.** *Reversible, with $P = Nk_BT/V$ throughout:*

$$W_{rev} = -\int_{V_i}^{V_f} \frac{Nk_BT}{V}\, dV = -Nk_BT \ln \frac{V_f}{V_i}.$$

Irreversible, in one step against $P_{ext} = P_f \neq P_{gas}$:

$$W_{irr} = -P_f(V_f - V_i).$$

Same end temperature, so $\Delta U = 0$ and $Q = -W$ in both, but $|W_{irr}| < |W_{rev}|$: the rectangle under P_f is smaller than the area under the isotherm.

Remark 15. *A reversible path gives the most work out of an expansion and needs the least work in for a compression. W and Q depend on the path; ΔU does not.*

5.2 Exact and Inexact Differentials

For a function $Z(x, y)$, a differential change has the form

$$dZ = M(x, y)\, dx + N(x, y)\, dy. \tag{5}$$

Theorem 1 (Test for exactness)**.** *dZ is exact (path independent, so Z is a state function) if and only if*

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y. \tag{6}$$

Otherwise dZ is inexact (path dependent).

Example (Internal energy of an ideal gas)**.** *For $U(V, T)$, since U depends only on T,*

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT = 0\, dV + C_V\, dT, \qquad C_V = \tfrac{f}{2} Nk_B.$$

Testing: $M = 0$ *and* $N = C_V$ *is constant, so*

$$\left(\frac{\partial M}{\partial T}\right)_V = 0 = \left(\frac{\partial N}{\partial V}\right)_T.$$

The differential is exact, so U *is a state function.*

Example (Work)**.** *For a reversible process,*

$$đW = -P\,dV + 0\,dT.$$

Testing: $M = -P = -nRT/V$ *and* $N = 0$*, so*

$$\left(\frac{\partial M}{\partial T}\right)_V = -\frac{nR}{V} \neq 0 = \left(\frac{\partial N}{\partial V}\right)_T.$$

The differential is inexact, so W *is not a state function.*

Remark 16. *Inexact differentials are written* $đQ$ *and* $đW$ *to signal that they depend on the path.*

Totals are integrals along the path:

$$W = \int đW = -\int_{V_1}^{V_2} P\,dV, \qquad Q = \int đQ,$$

so different paths between the same two states[7] give different W and Q, but the same $\Delta U = Q + W$.

5.3 Material Properties of a Compressible Fluid

A fluid has an equation of state $g(P, V, T) = 0$[8]. Picking two of (P, V, T) fixes the third, so we can write $V = V(P, T)$. By the chain rule, a small change in V is

$$dV = \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP,$$

where the subscript is the variable held fixed. These two derivatives are measurable properties of the material:

Definition 17 (Thermal expansivity, bulk modulus, compressibility)**.**

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P, \qquad B = -V\left(\frac{\partial P}{\partial V}\right)_T, \qquad \kappa = \frac{1}{B} = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T. \tag{7}$$

[7] Same start and end states, different route in between. A different end state is not a different path.

[8] for an ideal gas, $g = PV - Nk_BT$

In terms of these,

$$dV = \beta V\, dT - \frac{V}{B}\, dP. \tag{8}$$

Likewise, write dU two ways. By the chain rule (5), and by the first law (2) on a reversible step where $đW = -P\, dV$ from (4),

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT = đQ_{\text{rev}} - P\, dV.$$

These are equal as a whole, not term by term. Solving for the heat,

$$đQ_{\text{rev}} = \left(\frac{\partial U}{\partial T}\right)_V dT + \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right] dV. \tag{9}$$

At constant volume ($dV = 0$), $đQ_{\text{rev}} = (\partial U/\partial T)_V\, dT = C_V\, dT$.

Remark 17. *$(\partial U/\partial V)_T$ is not $-P$: that would be the change of U with V when no heat flows, not at constant T. For an ideal gas $(\partial U/\partial V)_T = 0$, since particles do not interact. For a real fluid it is nonzero and hard to measure directly; it will be recast into measurable form later using Maxwell relations.*

6 Heat Capacity

September 11, 2026

6.1 Definition

Definition 18 (Heat capacity)**.** *The heat capacity C of a system is the heat needed to raise its temperature by* 1 K*:*

$$C = \frac{Q}{\Delta T}. \tag{10}$$

Definition 19 (Specific heat capacity)**.** *The specific heat capacity $c \equiv C/m$ is intensive and a property of the material. With (10), the heat needed to change the temperature of a mass m by ΔT is*

$$Q = mc\, \Delta T. \tag{11}$$

6.2 Heat Capacity Depends on the Process

Using the first law, $Q = \Delta U - W$, so

$$C = \frac{\Delta U - W}{\Delta T}.$$

C depends on how much work is done during the heating (it would almost be better named the "energy capacity"), so the process must be specified.

Result 1 (Heat capacity at constant volume)**.** *If V is fixed, $W = 0$ and*

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V.$$

Result 2 (Heat capacity at constant pressure)**.** *Most materials expand when heated*[9]*, doing work* $W = -P\,\Delta V < 0$ *on the surroundings, so extra heat is needed. Dividing* (9) *by* dT *at constant* P*,*

$$C_P = C_V + \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right]\left(\frac{\partial V}{\partial T}\right)_P. \tag{12}$$

Remark 18. *For solids and liquids* $(\partial V/\partial T)_P \approx 0$*, so* $C_P \approx C_V$*; for gases the correction is significant.*

Proposition 1 (C_P of an ideal gas)**.** U *depends only on* T*, so* $(\partial U/\partial T)_P = C_V$*, and* $(\partial V/\partial T)_P = Nk_B/P$*. Hence*

$$C_P = C_V + Nk_B = C_V + nR, \qquad c_P = c_V + R \text{ (molar).}$$

6.3 Enthalpy

Definition 20 (Enthalpy)**.** *Enthalpy is the state function*

$$H = U + PV. \tag{13}$$

It is the convenient energy to use for processes at constant pressure.

Proposition 2. *At constant* P*,* $dH = đQ$*, and therefore*

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P. \tag{14}$$

Proof. Differentiate 13 with the product rule, $dH = dU + P\,dV + V\,dP$. On a reversible step, the first law (2) with $đW = -P\,dV$ from (4) gives $dU = đQ_{\text{rev}} - P\,dV$. Substituting,

$$dH = (đQ_{\text{rev}} - P\,dV) + P\,dV + V\,dP = đQ_{\text{rev}} + V\,dP.$$

At constant pressure $dP = 0$, so $dH = đQ_{\text{rev}}$. The heat capacity (10) is the heat absorbed per unit temperature change, so at constant pressure

$$C_P = \left(\frac{đQ_{\text{rev}}}{dT}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P.$$

□

6.4 Calorimetry

Definition 21 (Calorimeter)**.** *A calorimeter is an insulated container in which two bodies at different temperatures reach a common final temperature. Since the whole is isolated,*

$$Q_{system} = -Q_{surroundings}.$$

[9] Some materials contract over certain ranges of T (negative thermal expansion), but they are the exception.

Example. 150.0 g *of lead at* 100 °C *is dropped into* 50.0 g *of water at* 22.0 °C*; the final temperature is* 28.8 °C*. With* $c_{water} = 4.18\,\mathrm{J\,g^{-1}\,K^{-1}}$*,* (11) *gives*

$$q_{water} = (50\,\mathrm{g})(4.18\,\mathrm{J\,g^{-1}\,{}^\circ C^{-1}})(6.8\,{}^\circ\mathrm{C}) = 1.4\times10^3\,\mathrm{J} = -q_{lead},$$

$$c_{lead} = \frac{q_{lead}}{m_{lead}\,\Delta T_{lead}} = \frac{-1.4\times10^3\,\mathrm{J}}{(150\,\mathrm{g})(28.8-100)\,{}^\circ\mathrm{C}} = 0.13\,\mathrm{J\,g^{-1}\,{}^\circ C^{-1}}.$$

6.5 The Mechanical Equivalent of Heat

In Joule's experiment (1850), a falling weight spins a paddle wheel in insulated water; the lost potential energy $mg\,\Delta h$ raises the water's temperature. He found about 4.2 J per gram per °C, close to the modern $4.18\,\mathrm{J\,g^{-1}\,K^{-1}}$. This showed that W and Q are interchangeable forms of energy, and that a given amount of work always produces the same amount of heat (energy is conserved).

7 Quasistatic Processes of an Ideal Gas

September 9 and 14, 2026

Since $PV = Nk_BT$, how P changes with V depends on what happens to T, so $W = -\int P\,dV$ depends on the process. All processes below are quasistatic. Figure 1 shows them on a P-V diagram.

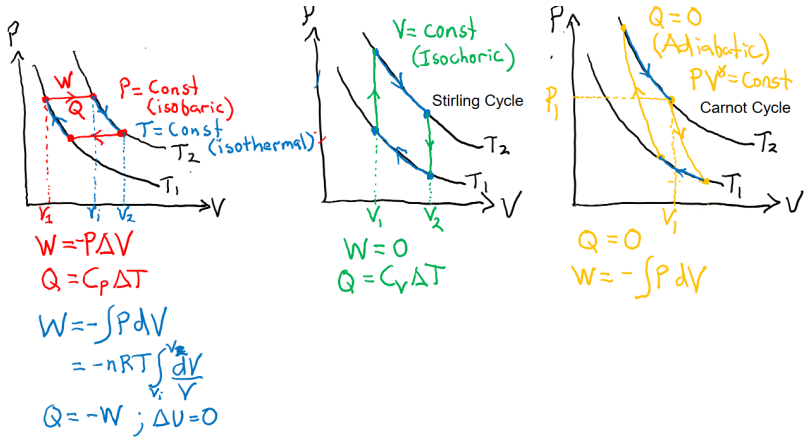


Figure 1: Left: isobaric (red) and isothermal (blue) processes between the isotherms T_1 and T_2. Centre: a Stirling cycle, made of two isochoric (green) and two isothermal steps. Right: a Carnot cycle, made of two adiabatic (yellow) and two isothermal steps.

7.1 Isobaric and Isochoric

Definition 22 (Isobaric process)**.** *A process at constant pressure.*

On a P-V diagram it is a horizontal line. With P constant, (4) and the definition of C_P give

$$W = -P(V_2 - V_1), \qquad Q = \int_{T_1}^{T_2} C_P \, dT = C_P \, \Delta T \quad \text{(constant } C_P\text{)}.$$

Definition 23 (Isochoric process)**.** *A process at constant volume.*

On a P-V diagram it is a vertical line. With $dV = 0$, (4) gives no work, so by the first law (2) all the heat goes into U:

$$W = 0, \qquad Q = \Delta U = C_V \, \Delta T.$$

7.2 Isothermal

Definition 24 (Isothermal process)**.** *A process at constant temperature.*

On a P-V diagram it follows the hyperbola $P = Nk_BT/V$, called an isotherm (Figure 2). The gas is in contact with a reservoir that supplies or absorbs heat. With $P = Nk_BT/V$,

$$W = -\int_{V_i}^{V_f} \frac{Nk_BT}{V} \, dV = Nk_BT \ln \frac{V_i}{V_f}.$$

Since U depends only on T (3), $\Delta U = \frac{f}{2} Nk_B(T_f - T_i) = 0$ and

$$Q = -W = Nk_BT \ln \frac{V_f}{V_i}.$$

Remark 19. *In an expansion the gas does work and absorbs an equal heat from the reservoir; in a compression the work done on the gas is dumped into the reservoir as heat.*

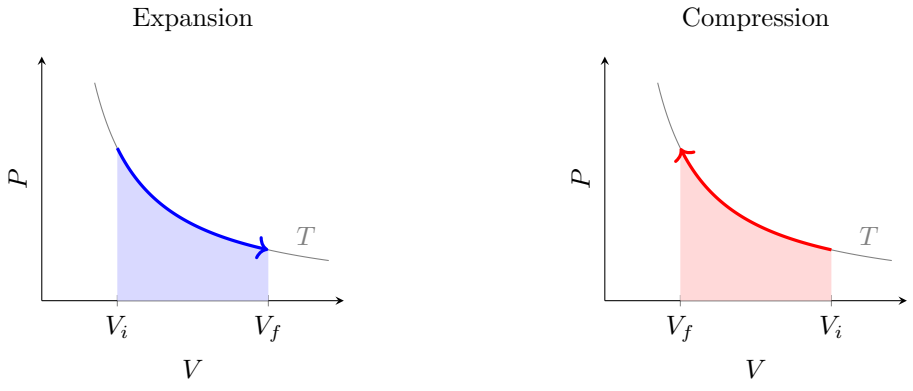


Figure 2: Isothermal processes of an ideal gas. The shaded area under the isotherm is $|W|$. Left: expansion, $W < 0$ and heat flows in. Right: compression, $W > 0$ and heat flows out.

7.3 Adiabatic

Definition 25 (Adiabatic process)**.** *A process with no heat flow, $Q = 0$.*

On a P-V diagram it follows an adiabat, $PV^\gamma = \text{constant}$ (Figure 3). The gas is insulated, or the process is too fast for heat to flow. The first law (2) gives $W = \Delta U = C_V\,\Delta T$: an adiabatic compression heats the gas and an expansion cools it.

Equation of the adiabat. With $Q = 0$, $dU = đW$. Using (3) for dU, (4) for $đW$, and $P = Nk_BT/V$,

$$\frac{f}{2}Nk_B\,dT = -\frac{Nk_BT}{V}\,dV \quad\Longrightarrow\quad \frac{f}{2}\frac{dT}{T} = -\frac{dV}{V} \quad\Longrightarrow\quad \frac{f}{2}\ln\frac{T_f}{T_i} = -\ln\frac{V_f}{V_i} \quad\Longrightarrow\quad VT^{f/2} = \text{constant}.$$

Proposition 3 (Adiabat in terms of P and V)**.**

$$PV^\gamma = \textit{constant}, \qquad TV^{\gamma-1} = \textit{constant}, \qquad \gamma = \frac{f+2}{f} = \frac{C_P}{C_V}.$$

Proof. Raise $VT^{f/2} = \text{constant}$ to the power $2/f$: $TV^{2/f} = \text{constant}$, and $2/f = \gamma - 1$. Substituting $T = PV/Nk_B$ gives $PV^{1+2/f} = PV^\gamma = \text{constant}$. Finally, $C_V = \frac{f}{2}Nk_B$ and $C_P = C_V + Nk_B = \frac{f+2}{2}Nk_B$, so $C_P/C_V = (f+2)/f = \gamma$. □

Remark 20. *Since $\gamma > 1$, an adiabat is steeper than an isotherm and crosses from one to a colder (expansion) or hotter (compression) one. Monatomic: $\gamma = 5/3$. Diatomic (air): $\gamma = 7/5$.*

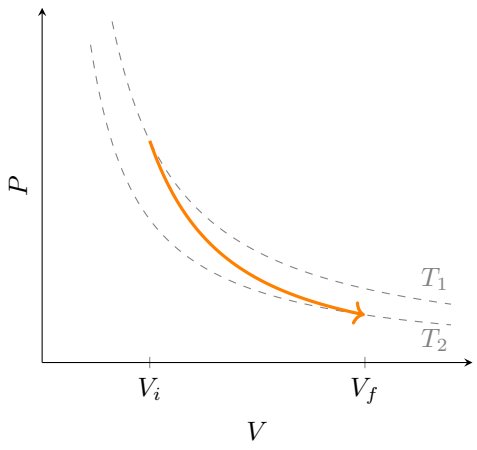


Figure 3: Adiabatic expansion (orange) between the isotherms $T_1 > T_2$ (dashed).

8 Kinetic Theory of Gases

September 14, 2026

Definition 26 (Kinetic theory)**.** *Kinetic theory is a microscopic model for $U(T)$ and C of an ideal gas: relate P to the kinetic energy of the particles, then use $PV = Nk_BT$ to relate that to T.*

8.1 Pressure from a Single Particle

One particle of mass m bounces between the walls of a box of length L with x-velocity u_x. Each collision reverses its momentum, giving the wall

$$\Delta p_{\text{wall}} = -(-mu_x - mu_x) = 2mu_x,$$

once every round trip $\Delta t = 2L/u_x$. The average force is

$$\bar{F} = \frac{\Delta p_{\text{wall}}}{\Delta t} = \frac{mu_x^2}{L} \qquad \text{(one particle)}.$$

8.2 Pressure from All N Particles

Summing over N particles and dividing by the wall area A ($AL = V$),

$$P = \frac{Nm\overline{u_x^2}}{V} \quad \Longrightarrow \quad PV = Nm\overline{u_x^2}.$$

8.3 Temperature and the Average Kinetic Energy

September 16, 2026

No direction is preferred, so $\overline{u^2} = 3\,\overline{u_x^2}$ and $PV = \frac{1}{3}Nm\overline{u^2}$. Comparing with $PV = Nk_BT$,

$$\bar{e}_k = \frac{1}{2}m\overline{u^2} = \frac{3}{2}k_BT.$$

Remark 21. *The average translational kinetic energy per molecule depends only on temperature, not on the mass or type of molecule. This is the microscopic meaning of temperature.*

Remark 22. *The speeds follow the Maxwell–Boltzmann distribution, with most probable speed $<$ average speed $<$ rms speed $\sqrt{\overline{u^2}}$.*

9 Equipartition and Heat Capacities

September 16, 2026

9.1 The Equipartition Theorem

Definition 27 (Degree of freedom)**.** *A degree of freedom (DOF) is a quadratic term in the total energy of a molecule. The number of such terms is f.*

Example (Monatomic gas)**.** *Translation: $\frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$, three terms (v_x, v_y, v_z), so $f = 3$.*

Principle 1 (Equipartition theorem)**.** *At temperature T, each degree of freedom has an average energy of $\frac{1}{2}k_BT$. So N molecules with f DOF each have*

$$U_{thermal} = Nf \cdot \frac{1}{2}k_BT.$$

Remark 23. *$U_{thermal}$ is not the total energy, which also includes fixed energy like chemical bonds. Equipartition still gives how U changes with T, unless bonds break or the phase changes.*

9.2 Counting Degrees of Freedom

Each of the N atoms in a molecule can move in x, y, z, giving $3N$ motions in total, split into:

- **Translational:** 3, the whole molecule moving.
- **Rotational:** 3 (nonlinear) or 2 (linear, since spinning about its own axis does nothing).
- **Vibrational:** the rest, $3N - 6$ (nonlinear) or $3N - 5$ (linear), atoms moving relative to each other.

Translations and rotations are kinetic only, so one DOF each. Each vibration is a spring, kinetic plus potential, so two DOF each.

Example (O_2, linear)**.** $3N = 6$*: 3 translational, 2 rotational,* $6 - 5 = 1$ *vibration, so* $f = 3 + 2 + 2(1) = 7$.

Example (H_2O, nonlinear)**.** $3N = 9$*: 3 translational, 3 rotational,* $9 - 6 = 3$ *vibrations, so* $f = 3 + 3 + 2(3) = 12$.

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Solids. Atoms are pinned near lattice sites, so only vibration remains. With n atoms per unit cell there are $3n$ vibrational modes, so $f = 6n$ ($f = 6$ for a simple solid).

9.3 Freezing Out

Each DOF has a minimum quantum excitation energy (for a vibration, $\hbar\omega_{\text{vib}}$). If $\frac{1}{2}k_BT$ is much smaller than it, the DOF stays in its ground state and is frozen out. So $f \to f(T)$ and $C_V \to C_V(T)$.

9.4 Predicting Heat Capacities

From $U = \frac{f}{2}Nk_BT$,

$$C_V = \frac{f}{2}Nk_B.$$

Example (Monatomic gas)**.** $f = 3$*, which cannot freeze out:* $c_V = \frac{3}{2}R \approx 12.5\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$*, independent of the atom and of* T *(*$R = 8.31\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$*).*

Example (Diatomic H_2). *$c_V = \frac{3}{2}R$ at low T, $\frac{5}{2}R$ once rotation turns on (above ~ 100 K), and $\frac{7}{2}R$ once vibration turns on ($k_B T \sim \hbar\omega_{vib}$, thousands of K).*

Proposition 4 (Rule of Dulong and Petit). *At high T, with all vibrational modes active, a monatomic solid ($f = 6$) has*

$$c_V \approx 3R \approx 25\,\mathrm{J\,mol^{-1}\,K^{-1}}.$$

10 Heat Engines and the Second Law

Remark 24. *Convention for this section: T_1 is the hot reservoir and T_2 the cold one, and Q_1, Q_2, W are magnitudes (all positive).*

10.1 Cycles and Heat Engines

Definition 28 (Cycle). *A process that returns the system (working substance) to its initial state. Then $\Delta U = 0$, and the net work is the area enclosed by the loop on the P-V diagram.*

Definition 29 (Heat engine). *A device that takes a working substance around a cycle, absorbing Q_1 from a hot reservoir at T_1, rejecting Q_2 to a cold one at T_2, and producing work W. By the first law, $W = Q_1 - Q_2$, and its efficiency is*

$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1}.$$

Remark 25. *Getting $\eta = 1$ would need $Q_2 = 0$. The second law forbids this. Any heat taken from the hot reservoir and not turned into work is a loss.*

- Stirling cycle: two isotherms joined by two isochores.
- Carnot cycle: two isotherms joined by two adiabats.

Example (Nuclear plant). *$\eta = 0.31$, electrical output $625\,\mathrm{MW}$. Per day, $W = 625 \times 10^6\,\mathrm{W} \times 86\,400\,\mathrm{s} = 5.4 \times 10^{13}\,\mathrm{J}$, so $Q_1 = W/\eta = 1.74 \times 10^{14}\,\mathrm{J}$ and the waste heat is $Q_2 = Q_1 - W = 1.20 \times 10^{14}\,\mathrm{J}$, which is why plants need cooling towers.*

10.2 Statements of the Second Law

Law 4 (Kelvin–Planck statement). *It is impossible to construct a device that, operating in a cycle, produces no effect other than the extraction of heat from a single body at uniform temperature and the production of an equivalent amount of work.*

Law 5 (Clausius statement). *It is impossible to construct a device that, operating in a cycle, produces no effect other than the transfer of heat from a colder to a hotter body.*

Remark 26. *The key words matter:*

- *Cycle: an isothermal expansion of an ideal gas converts heat entirely into work, but it is not a cycle.*
- *No other effect: the reservoirs only deliver heat, no work.*
- *Single body: there is no reservoir to accept waste heat.*

The converse is allowed: converting work entirely into heat is possible. A refrigerator or heat pump does not violate Clausius, since work is done on it.

Proposition 5. *The Kelvin and Clausius statements are equivalent.*

Proof. If Kelvin is false, an engine E turns Q_1 from the hot body entirely into $W = Q_1$. Use W to drive a refrigerator R that takes Q_2 from the cold body and delivers $Q_1 + Q_2$ to the hot body. The composite moves Q_2 from cold to hot with no other effect, violating Clausius.

If Clausius is false, a device moves Q_2 from cold to hot with no work. Run an ordinary engine E that takes Q_1 from the hot body, rejects Q_2 to the cold body and produces $W = Q_1 - Q_2$. The cold body is unchanged, so the composite turns $Q_1 - Q_2$ from the hot body entirely into work, violating Kelvin. □

10.3 The Carnot Cycle and Carnot's Theorem

Definition 30 (Carnot cycle)**.** *A fully reversible cycle between only two reservoirs:*

1. *isothermal expansion at T_1, absorbing Q_1;*
2. *adiabatic expansion, $T_1 \to T_2$;*
3. *isothermal compression at T_2, rejecting Q_2;*
4. *adiabatic compression, $T_2 \to T_1$.*

Theorem 2 (Carnot's theorem)**.** *No engine operating between two reservoirs is more efficient than a Carnot engine operating between the same two reservoirs.*

Proof. Suppose an engine E' has $\eta' > \eta_C$. Scale the cycles so that $W' = W$; then $W/Q_1' > W/Q_1$ gives $Q_1 > Q_1'$. Since the Carnot engine C is reversible, run it backwards as a refrigerator driven by the work of E'. The composite does no net work, but delivers $Q_1 - Q_1' > 0$ to the hot reservoir and takes the same amount from the cold one, violating Clausius. □

Corollary 2.1. *All Carnot (reversible) engines between the same two reservoirs have the same efficiency, independent of the working substance. It depends only on the reservoir temperatures.*

10.4 Thermodynamic Temperature Scale

Definition 31 (Thermodynamic temperature)**.** *Since η_C depends only on the reservoirs, define the thermodynamic temperature by*

$$\eta_C = 1 - \frac{T_2}{T_1}, \qquad \text{so} \qquad \frac{T_1}{T_2} = \frac{Q_1}{Q_2} \quad \text{(Carnot only).}$$

Remark 27. *The scale is self-consistent: stacking Carnot engines C_{12} and C_{23} on a ladder $T_1 > T_2 > T_3$ gives $T_1/T_3 = Q_1/Q_3$, without T_2. It is independent of the working substance (heat can always be measured), and it coincides with the ideal gas scale up to a constant factor.*

10.5 Refrigerators and Heat Pumps

A Carnot engine run backwards takes Q_2 from the cold reservoir, uses work $W = Q_1 - Q_2$, and delivers Q_1 to the hot one. Performance is measured by the coefficient of performance (COP), which can exceed 1.

Proposition 6 (Carnot COP)**.**

$$\textit{Refrigerator (output } Q_2\textit{):} \quad COP = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2},$$

$$\textit{Heat pump (output } Q_1\textit{):} \quad COP = \frac{Q_1}{W} = \frac{T_1}{T_1 - T_2} = \frac{1}{1 - T_2/T_1}.$$

Example. *Fridge compartment at* $276\,\mathrm{K}$ *in a* $293\,\mathrm{K}$ *room: maximum COP* ≈ 16*. Freezer at* $255\,\mathrm{K}$*: only* ≈ 6.7*. Real fridges: about 4 or 5. Heat pump from* $273\,\mathrm{K}$ *outside to* $293\,\mathrm{K}$ *inside: COP* ≈ 15*.*

Remark 28. *The COP of both drops as $T_1 - T_2$ grows (as T_2/T_1 decreases).*

10.6 Real Engines

Example (Steam engine)**.** *Between* $773\,\mathrm{K}$ *and* $293\,\mathrm{K}$*, $\eta_C = 1 - 293/773 = 0.62$, but real steam engines reach about* 0.4*.*

Remark 29. *Real engines do not run Carnot cycles: setting up isothermal and adiabatic steps is impractical, isothermal heat flow is far too slow to deliver useful power, and hot parts always leak heat to the environment. $\eta_C = 1$ would need $T_2 = 0$, forbidden by the third law.*

Otto cycle. A simplified model of a gasoline engine (a good model of the 2-stroke cycle, or of the compression/expansion part of the 4-stroke cycle). The gas is taken to be air, an ideal diatomic gas with $\gamma = 7/5$ (the mixture is only $\sim 6\%$ fuel):

- $a \to b$: adiabatic compression, $V_1 \to V_2$ (fast, so little heat flows);

- $b \to c$: ignition at constant V_2, $Q_1 = C_V(T_c - T_b)$;
- $c \to d$: adiabatic expansion, $V_2 \to V_1$;
- $d \to a$: exhaust at constant V_1, $Q_2 = C_V(T_d - T_a)$.

Proposition 7 (Otto efficiency)**.** *With compression ratio $r_c = V_1/V_2$,*

$$\eta = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1} = 1 - \frac{1}{r_c^{\gamma-1}}.$$

Proof. Along the adiabats $TV^{\gamma-1}$ is constant: $T_a V_1^{\gamma-1} = T_b V_2^{\gamma-1}$ and $T_d V_1^{\gamma-1} = T_c V_2^{\gamma-1}$. Subtracting, $T_d - T_a = (T_c - T_b)(V_2/V_1)^{\gamma-1}$, so

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_d - T_a}{T_c - T_b} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}.$$

□

Remark 30. *Efficiency improves with r_c. Cars have $r_c \approx 7$ or 8, giving $\eta \approx 1 - 7^{-2/5} = 0.54$, while real engines reach about 30%. High-performance cars have $r_c \geq 10$ (F1 capped r_c at 16 for 2026). Schroeder 4.3 also covers the Diesel, Stirling and Rankine (steam) cycles.*

11 Entropy

The second law is about the direction of spontaneous processes: heat flows from hot to cold, and the reverse requires work. Entropy quantifies this. The plan: derive the Clausius inequality for any cycle, use it on reversible paths to define S as a state function, then compare reversible and irreversible processes.[10]

11.1 The Clausius Inequality

Theorem 3 (Clausius inequality)**.** *For any cycle,*

$$\oint \frac{đQ}{T_{surr}} \leq 0,$$

where T_{surr} is the temperature of the reservoir supplying heat at each point of the cycle (not T_{sys}).

Proof. Supply each small heat $đQ_i$ to the working substance from an auxiliary reservoir at T_i, kept unchanged by a Carnot engine C_i running between it and a reservoir at T_0. That reservoir is in turn refilled from a single principal reservoir

[10] Schroeder does not follow this approach; the closest is 3.1 and 3.2 (Measuring Entropy, The Macroscopic View of Entropy).

at T_0, which therefore gives $(T_0/T_i)đQ_i$ while C_i does work $đW_i$. Take the composite system (working substance, Carnot engines, auxiliary reservoirs). After one full cycle it is unchanged, so $\Delta U = 0$ and the external work done by it is

$$W_{\text{ext}} = Q = T_0 \sum_i \frac{đQ_i}{T_i}.$$

If $W_{\text{ext}} > 0$, heat from a single reservoir became work in a cycle, violating Kelvin. So $W_{\text{ext}} = Q \le 0$, i.e. $\sum_i đQ_i/T_i \le 0$, which becomes the integral in the limit. □

Corollary 3.1. *For a fully reversible cycle, running it backwards flips the sign of every $đQ$, so both $\le$ and $\ge$ hold:*

$$\oint_R \frac{đQ_R}{T} = 0.$$

The subscript "surr" is dropped since $T_{sys} = T_{surr}$ on a reversible path. If $T_{sys} \neq T_{surr}$ anywhere, the process is irreversible.

11.2 Definition of Entropy

Take a reversible cycle: $i \to f$ along R_1, back along R_2. The corollary gives $\int_{R_1} đQ_R/T = -\int_{R_2, f\to i} đQ_R/T = \int_{R_2, i\to f} đQ_R/T$, so the integral is the same along every reversible path.

Definition 32 (Entropy)**.** *Entropy S is the state function with*

$$\Delta S = S_f - S_i = \int_{i,\,R}^{f} \frac{đQ_R}{T}, \qquad dS = \frac{đQ_R}{T} \quad \textit{(reversible only).}$$

Remark 31. *Only differences in S are defined here. The integral must be along a reversible path ($T_{sys} = T_{surr}$), with T in kelvin; S is in $\mathrm{J\,K^{-1}}$. The same $đQ$ gives a smaller dS at higher T.*

Remark 32. *For an irreversible process between two equilibrium states, ΔS is still defined: pick any convenient reversible path between the same endpoints.*

Example (Heating water)**.** $1\,\mathrm{kg}$ *of water ($C_P = 4190\,\mathrm{J\,K^{-1}}$) is heated from* $20\,^\circ\mathrm{C}$ *to* $100\,^\circ\mathrm{C}$ *on a stove at* $100\,^\circ\mathrm{C}$*. This is irreversible (large T difference), so use a reversible path through reservoirs at $T + dT$: $đQ = C_P\,dT$,*

$$\Delta S = C_P \int_{T_i}^{T_f} \frac{dT}{T} = C_P \ln\frac{T_f}{T_i} = (4190\,\mathrm{J\,K^{-1}}) \ln\frac{373}{293} = 1.01 \times 10^3\,\mathrm{J\,K^{-1}}.$$

Example (Free expansion)**.** *An ideal gas expands freely into vacuum, doubling its volume. $Q = W = 0$, so $\Delta U = 0$ and T is unchanged. Use a reversible isothermal expansion between the same states: $Q_R = Nk_BT\ln 2$, so*

$$\Delta S = \frac{Q_R}{T} = Nk_B \ln 2 > 0,$$

even though no heat flowed in the actual (adiabatic but irreversible) process.

11.3 Principle of Increasing Entropy

Theorem 4. *For any infinitesimal process between equilibrium states,*

$$dS \geq \frac{đQ}{T_{surr}},$$

with equality if and only if the process is reversible.

Proof. Build a cycle: $i \to f$ along the actual (possibly irreversible) path, then back $f \to i$ along a reversible path R. Clausius gives

$$\int_i^f \frac{đQ}{T_0} + \int_{f,\,R}^{i} \frac{đQ_R}{T} \leq 0 \quad \Longrightarrow \quad \int_i^f \frac{đQ}{T_0} \leq \int_{i,\,R}^{f} \frac{đQ_R}{T} = S_f - S_i.$$

□

Corollary 4.1 (Principle of increasing entropy)**.** *For a thermally isolated system, $đQ = 0$, so*

$$\Delta S \geq 0.$$

The entropy of a thermally isolated system increases in any irreversible process and is unchanged in any reversible one. Isolated systems evolve toward the equilibrium state of maximum entropy.

Remark 33. *If the system is also mechanically isolated (no work), U is constant too.*

Remark 34. *The principle applies to the whole isolated system, not to its parts. If $T_A > T_B$ inside an adiabatic wall, heat flows from A to B: S_A decreases, S_B increases, and the total increases.*

Remark 35 (Entropy maximization)**.** *A thermally isolated system evolves until S is maximal; a totally isolated one (thermally and mechanically) until S is maximal at fixed U. For a thermodynamic universe made of such systems, $\Delta S_{universe} \geq 0$. Whether this applies to the actual universe depends on whether it is finite and isolated.*

Remark 36 (Arrow of time)**.** *Most dynamical laws (Newton's second law, the Schrödinger equation) are unchanged under $t \to -t$. Entropy increase is one of the few laws that picks a direction, and it is the most widely accepted explanation of the arrow of time.*

11.4 Temperature-Entropy Diagrams

Any two independent state variables fix the state, so (T, S) works as well as (P, V). On a reversible path $đQ_R = T\,dS$, so

$$Q = \int_R T\,dS,$$

the area under the curve on a T-S diagram, just as $|W|$ is the area under a P-V curve.

Definition 33 (Isentropic process)**.** *A process at constant S. Since $dS = đQ_R/T$, a process is isentropic if and only if it is both adiabatic and reversible. An irreversible adiabatic process, like a free expansion, is not isentropic.*

Example (Carnot cycle on a T-S diagram)**.** *The isotherms are horizontal and the reversible adiabats are vertical (isentropes), so the cycle is a rectangle (Figure 4). With ΔS its width, $Q_1 = T_1\,\Delta S$ and $Q_2 = T_2\,\Delta S$, so the net heat, equal to the net work, is the enclosed area, and*

$$\eta_C = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}.$$

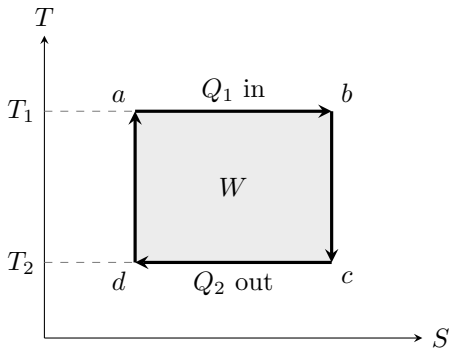


Figure 4: Carnot cycle on a T-S diagram: $a \to b$ and $c \to d$ isothermal, $b \to c$ and $d \to a$ isentropic. The enclosed area is $Q_1 - Q_2 = W$.

11.5 The Thermodynamic Identity

Theorem 5 (Thermodynamic identity)**.**

$$dU = T\,dS - P\,dV, \qquad \text{or} \qquad T\,dS = dU + P\,dV. \tag{15}$$

Proof. On a reversible step, the first law (2) reads $dU = đQ_{\text{rev}} + đW_{\text{rev}}$, with $đW_{\text{rev}} = -P\,dV$ from (4) and $đQ_{\text{rev}} = T\,dS$ from the definition of entropy. □

Remark 37. *It was derived on a reversible step, but it contains only state variables (T, P) and state functions (U, S, V), so it holds for any infinitesimal process between equilibrium states. For an irreversible one, $T\,dS$ and $-P\,dV$ are no longer $đQ$ and $đW$ separately. It combines the first and second laws, and the rest of thermodynamics follows from it the way mechanics follows from Newton's laws.*

11.6 Entropy of an Ideal Gas

Proposition 8 (Molar entropy of an ideal gas)**.** *With molar quantities $s = S/n$, $c_v = C_V/n$ and $v = V/n$,*

$$s = c_v \ln T + R \ln v + s_0,$$

where s_0 is a constant fixed by a reference state (T_0, v_0).

Proof. For an ideal gas $U = U(T)$, so $dU = C_V\,dT$, and $P = nRT/V$. The thermodynamic identity (15) gives

$$T\,dS = C_V\,dT + \frac{nRT}{V}\,dV.$$

Dividing by nT, $ds = c_v\,dT/T + R\,dv/v$. Integrating from (T_0, v_0) to (T, v) and absorbing $-c_v \ln T_0 - R \ln v_0$ into s_0 gives the result. □

Remark 38. *s_0 cancels in any difference $s_f - s_i$, which is all classical thermodynamics defines.*

Example (Free expansion again)**.** *At constant T, doubling the volume gives $\Delta s = R \ln 2$. For $1\,\text{mol}$, $\Delta S = (8.315\,\text{J mol}^{-1}\,\text{K}^{-1}) \ln 2 = 5.76\,\text{J K}^{-1}$, the same as the free expansion result, since S is a state function.*

11.7 Entropy of the Universe

For any process, $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$.

Example (Heating water on one reservoir)**.** *In the water example, the reservoir stays at $T_f = 373\,\text{K}$ and loses $Q = C_P(T_f - T_i)$. Computing its entropy change along a reversible path (as if through an intermediate reservoir just below $100\,^\circ\text{C}$),*

$$\Delta S_{reservoir} = -\frac{Q}{T_f} = -C_P\,\frac{80\,\text{K}}{373\,\text{K}},$$

$$\Delta S_{universe} = C_P\left[\ln\frac{373}{293} - \frac{80}{373}\right] = C_P(0.241 - 0.214) = 0.027\,C_P > 0.$$

Remark 39. *$\Delta S_{universe}$ is much smaller than the changes of either part, but no choice of parameters can make it negative.*

Example (Heating water in two stages)**.** *First on a reservoir at $323\,\text{K}$, then on one at $373\,\text{K}$. ΔS_{water} is unchanged (same end states), and*

$$\Delta S_{universe} = C_P\left[\ln\frac{373}{293} - \frac{30}{323} - \frac{50}{373}\right] = 0.014\,C_P,$$

about half the one-stage result. More stages bring the process closer to reversible.

Proposition 9 (Reversible heating)**.** *If the water is heated by an infinitesimal Carnot engine running between the reservoir at $373\,\text{K}$ and the water at T, then $\Delta S_{universe} = 0$.*

Proof. In each tiny cycle the engine takes $đQ_1$ from the reservoir and delivers $đQ_2 = C_P\,dT$ to the water. For a Carnot engine $đQ_1/đQ_2 = (373\,\text{K})/T$, so

$$\Delta S_{\text{reservoir}} = -\int \frac{đQ_1}{373\,\text{K}} = -\int_{293}^{373} \frac{C_P\,dT}{T} = -C_P \ln\frac{373}{293} = -\Delta S_{\text{water}}.$$

The engine returns to its initial state every cycle, so $\Delta S_{\text{universe}} = 0$. □

12 Statistical Thermodynamics

Classical thermodynamics says entropy increases in irreversible processes, but not what S measures. The statistical approach answers this by counting the ways energy can be arranged among the constituents of a system (combinatorics). The conclusion: irreversible processes are not inevitable, only overwhelmingly probable for macroscopic systems. This is Schroeder chapter 2.

12.1 Microstates and Macrostates

Flip three distinguishable coins (penny, nickel, dime). Each has 2 outcomes, so there are $2^3 = 8$ equally likely outcomes. If only the number of heads can be measured, there are 4 possible results, with probabilities 1/8 (0 heads), 3/8 (1), 3/8 (2), 1/8 (3).

Definition 34 (Microstate, macrostate, multiplicity)**.** *A microstate specifies the state of every individual particle (e.g. HHT). A macrostate specifies only a coarse, measurable quantity (e.g. two heads). The multiplicity Ω of a macrostate is the number of microstates giving that macrostate.*

Remark 40. *Macrostates can be told apart by measurement; microstates with the same macrostate cannot. Knowing the microstate fixes the macrostate, not the reverse.*

Example (Three coins)**.** $\Omega(3) = 1$, $\Omega(2) = 3$, $\Omega(1) = 3$, $\Omega(0) = 1$*: the $n = 3$ row of Pascal's triangle. The total is $\Omega(all) = 8$.*

Proposition 10 (Multiplicity of N coins)**.** *The probability of a macrostate is $\Omega(n)/\Omega(all)$, and for N coins with n heads,*

$$\Omega(N, n) = \binom{N}{n} = \frac{N!}{n!\,(N-n)!}, \qquad \Omega(all) = 2^N.$$

Remark 41. *There are only $N + 1$ macrostates but 2^N microstates.*

12.2 Sharpening of the Distribution

The number of heads follows a binomial distribution with $p = 1/2$: mean $N/2$, variance $N/4$, so $\sigma = \sqrt{N}/2$ and

$$\frac{\sigma}{N} = \frac{1}{2\sqrt{N}}.$$

The fraction of heads is peaked ever more sharply at $1/2$ as N grows. A fraction of 0.4 is no surprise with 10 coins, but with 1000 coins it is more than 6σ away and essentially never happens.

12.3 The Two-State Paramagnet

Definition 35 (Two-state paramagnet)**.** *A material of N magnetic dipoles (unpaired electrons or nuclear moments) that do not interact with each other. Along an applied $\vec{B}$, quantum mechanics allows each dipole only two states: up (aligned) or down (anti-aligned).*

The combinatorics are those of coins. The macrostate is labelled by $N_\uparrow$, with magnetization $M \propto N_\uparrow - N_\downarrow$, and

$$\Omega(N_\uparrow) = \binom{N}{N_\uparrow} = \frac{N!}{N_\uparrow!\, N_\downarrow!}.$$

Remark 42. *For $\vec{B} \approx \vec{0}$ each dipole is a fair coin, so we measure $M \approx 0$, simply because it is the overwhelmingly most probable macrostate.*

12.4 The Einstein Solid

Definition 36 (Einstein solid)**.** *A collection of N identical quantum harmonic oscillators with frequency f. Each has evenly spaced levels separated by hf, so the total energy is $U = q\,hf$ with q the number of energy units. In 3D each atom oscillates in 3 directions, so N oscillators are $N/3$ atoms.*

Proposition 11 (Multiplicity of an Einstein solid)**.**

$$\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q!\,(N - 1)!}.$$

Proof. Represent a microstate by q dots (energy units) and $N-1$ bars (partitions between oscillators). For example, with $N = 4$, $\bullet\,|\,\bullet\bullet\bullet\,||\,\bullet\bullet\bullet\bullet$ gives energies $1, 3, 0, 4$. Every arrangement of the $q + N - 1$ symbols is a distinct microstate, and there are $\binom{q+N-1}{q}$ ways to choose which are dots. □

Example ($N = 3$)**.** $\Omega(0) = 1$, $\Omega(1) = 3$, $\Omega(2) = 6$, $\Omega(3) = 10$.

12.5 Two Einstein Solids in Thermal Contact

Take two solids A and B that exchange energy inside an isolated box, each internally in equilibrium (local equilibrium). The macrostate is labelled by q_A, with $q_B = q - q_A$. The solids are independent, so for each microstate of A there are Ω_B of B:

$$\Omega_{\text{total}} = \Omega_A\,\Omega_B.$$

Example ($N_A = N_B = 3$, $q = 6$)**.**

q_A	Ω_A	q_B	Ω_B	Ω_{total}
0	*1*	*6*	*28*	*28*
1	*3*	*5*	*21*	*63*
2	*6*	*4*	*15*	*90*
3	*10*	*3*	*10*	*100*
4	*15*	*2*	*6*	*90*
5	*21*	*1*	*3*	*63*
6	*28*	*0*	*1*	*28*
				$462 = \binom{6+6-1}{6}$

Principle 2 (Fundamental assumption of statistical mechanics)**.** *In an isolated system in thermal equilibrium, all accessible microstates are equally probable.*

Remark 43. *Microscopically every process is reversible: the system moves freely between microstates. But macrostates are not equally probable. Above,* $q_A = 3$ *has probability* $100/462$ *and* $q_A = 0$ *only* $28/462$*. So energy starting all in B will probably spread out, though with numbers this small, finding it all in B again is not surprising.*

Example ($N_A = 300$, $N_B = 200$, $q = 100$)**.** *The peak is at* $q_A = 60$ *(energy shared in proportion to the number of oscillators), with* $\Omega_{total}(60)/\Omega_{total}(0) \sim 10^{33}$*. Nearly all of* $\Omega(all) \approx 9.3 \times 10^{115}$ *comes from* $q_A \approx 55$ *to* 65*. The probability of* $q_A < 30$ *or* $q_A > 90$ *is below* 10^{-6}*, and of* $q_A < 10$ *below* 10^{-20}*: checking a hundred times per second for the age of the universe (*$< 10^{18}$ *s) would barely catch it once.*

Remark 44 (Irreversibility)**.** *Starting at* $q_A = 0$ *or* 40 *and waiting, we will almost certainly find energy has flowed from B to A toward* $q_A \approx 60$*, and stays there up to small fluctuations. Heat flow is a probabilistic phenomenon: not certain, just overwhelmingly likely. This "law of increase of multiplicity" is a statistical form of the second law. It is not fundamental, only a very strong statement about probabilities, which becomes a law as N grows.*

12.6 Large Systems

For $N \sim 10^{23}$ we cannot compute factorials directly.

Proposition 12 (Stirling's approximation)**.** *For* $N \gg 1$*,*

$$N! \approx N^N e^{-N} \sqrt{2\pi N}, \qquad \ln N! \approx N \ln N - N.$$

Remark 45. *Derived in Schroeder Appendix B. For* $N = 100$ *the first is off by* 0.083% *and the second by* 0.89%*.*

Proposition 13 (Large Einstein solid, high temperature)**.** *For* $q \gg N$*,*

$$\Omega(N, q) \approx \left(\frac{eq}{N}\right)^N.$$

Proof. Replacing $(N-1)!$ by $N!$ and $(q+N-1)!$ by $(q+N)!$ changes Ω only by factors of order q and N, negligible against a number this large, so $\Omega \approx (q+N)!/(q!\,N!)$. By Stirling,

$$\ln \Omega \approx (q+N)\ln(q+N) - q\ln q - N\ln N.$$

For $q \gg N$, $\ln(q+N) = \ln q + \ln(1+N/q) \approx \ln q + N/q$, so

$$\ln \Omega \approx N\ln\frac{q}{N} + N + \frac{N^2}{q} \approx N\ln\frac{q}{N} + N.$$

Exponentiating gives the result. □

Remark 46. *Since the exponent N is huge, a small increase in N or q increases Ω enormously.*

Proposition 14 (Width of the peak)**.** *For two solids with $N_A = N_B = N$ and $q \gg N$, writing $q_A = q/2 + x$, $q_B = q/2 - x$ with $x \ll q$,*

$$\Omega \approx \Omega_{\max}\, e^{-N(2x/q)^2}, \qquad \Omega_{\max} = \left(\frac{e}{N}\right)^{2N}\left(\frac{q}{2}\right)^{2N}.$$

This is a Gaussian of width $\sigma = q/(2\sqrt{N})$.

Proof. $\Omega = (e/N)^{2N}(q_A q_B)^N = (e/N)^{2N}\left[(q/2)^2 - x^2\right]^N$, maximal at $x = 0$. Taking the log,

$$N\ln\left[\left(\tfrac{q}{2}\right)^2 - x^2\right] = N\ln\left(\tfrac{q}{2}\right)^2 + N\ln\left[1 - \left(\tfrac{2x}{q}\right)^2\right] \approx N\ln\left(\tfrac{q}{2}\right)^2 - N\left(\tfrac{2x}{q}\right)^2.$$

Ω falls to $1/e$ of $\Omega_{\max}$ when $N(2x/q)^2 = 1$, i.e. $x = q/(2\sqrt{N})$. □

Definition 37 (Thermodynamic limit)**.** *The limit where a system is large enough that measurable fluctuations away from the most likely macrostate never occur.*

Example. *For $N = 10^{20}$, $\sigma/(q/2) = 1/\sqrt{N} = 10^{-10}$: seeing a one-$\sigma$ fluctuation would need the energy of a subsystem to ten significant figures. So once equilibrium is reached, we may take the system to be in its most likely macrostate.*

12.7 Statistical Definition of Entropy

Any large system in equilibrium is found in the macrostate of greatest multiplicity, up to fluctuations too small to measure. The statistical second law: multiplicity tends to increase.

Definition 38 (Boltzmann entropy)**.**

$$S \equiv k_B \ln \Omega.$$

Remark 47. *The log tames the huge numbers and is monotonic, so it changes nothing qualitatively; k_B gives S units of $\mathrm{J\,K^{-1}}$, matching the classical state function. Entropy measures multiplicity.*

Example (Large Einstein solid)**.** *With $\Omega = (eq/N)^N$,*

$$S = k_B \ln \left(\frac{eq}{N}\right)^N = N k_B \left[\ln \frac{q}{N} + 1\right].$$

For $N = 10^{22}$ and $q = 10^{24}$, $S = N k_B (5.6) = 0.77\,\mathrm{J\,K^{-1}}$.

Remark 48. *S increases with both q and N, though not in proportion. Unlike the classical definition, $S = k_B \ln \Omega$ gives absolute values, not just differences.*

Proposition 15 (Additivity)**.** *For two weakly coupled, independent systems in the thermodynamic limit,*

$$S_{total} = S_A + S_B.$$

Proof. Rigorously $\Omega_{\text{total}} = \sum_{A,B} \Omega_A \Omega_B$ over all macrostates allowed by the total energy. In the thermodynamic limit the sum is dominated by the peak, $\Omega_{\text{total}} \approx \Omega_{A,\max}\, \Omega_{B,\max}$, so

$$S_{\text{total}} = k_B \ln \left(\Omega_{A,\max}\, \Omega_{B,\max}\right) = k_B \ln \Omega_{A,\max} + k_B \ln \Omega_{B,\max} = S_A + S_B.$$

□

Remark 49. *For small N the macrostates of A and B fluctuate and the full sum is needed.*

Law 6 (Second law, statistical form)**.** *Spontaneous, irreversible processes increase the entropy (total multiplicity) of the universe. Any large system in equilibrium is found in the macrostate of greatest entropy.*

Remark 50 (Entropy is not disorder)**.** *"Entropy = disorder" works for a shuffled deck (sorted has $\Omega = 1$), but fails elsewhere: crushed ice looks more disordered than water, yet water has much higher entropy. Think "entropy = multiplicity" instead.*

12.8 Maxwell's Demon

Maxwell (1867) imagined a demon working a door between two chambers of gas, letting only fast molecules through one way and slow ones the other. One side warms, the other cools, and total entropy seems to drop with no work done. This led to links between entropy and information theory (Landauer). The consensus is that the demon cannot violate the second law once the entropy it generates itself is counted.

12.9 Entropy of an Ideal Gas

Reading: Schroeder 2.5 and 2.6. The multiplicity of a monatomic ideal gas is

$$\Omega_N \approx \frac{1}{N!}\frac{V^N}{h^{3N}}\frac{\pi^{3N/2}}{(3N/2)!}\left(\sqrt{2mU}\right)^{3N} = f(N)\,V^N U^{3N/2},$$

which leads to the Sackur–Tetrode equation

$$S = Nk_B\left[\ln\left(\frac{V}{N}\left(\frac{4\pi mU}{3Nh^2}\right)^{3/2}\right) + \frac{5}{2}\right].$$

The text also covers interacting gases and the entropy of mixing.

A Appendix

B Useful Links